

What is an LLM?

In one boardroom sentence.

A plain-English tour of the architecture behind GPT, Claude and Gemini — and why it matters for AI delivery.

**"Weak quarterly results led the board
to replace the _____"**

A good model predicts: "CEO"

It has never seen your company. It just predicts the next word.

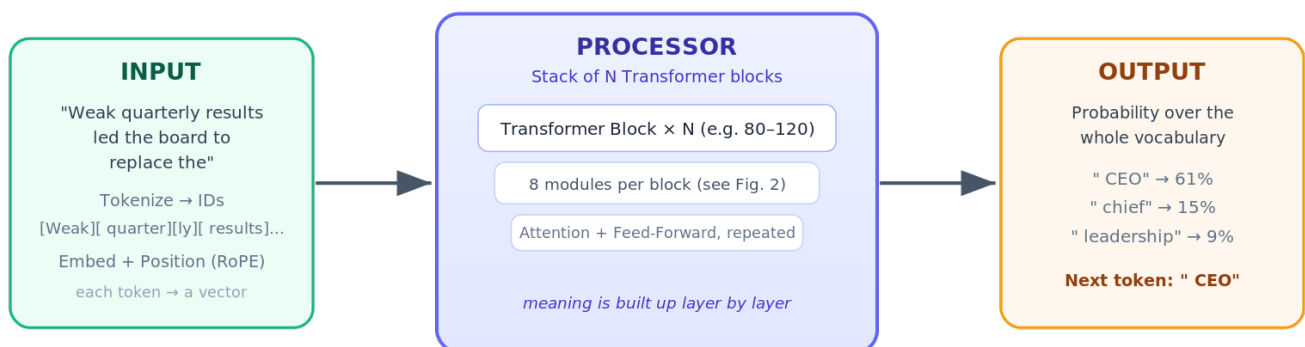
Swipe →

Input → Processor → Output, in 8 slides.

THE BIG PICTURE

Three stages, one engine

The Big Picture: Input → Processor → Output



The chosen token is appended to the input and the loop repeats — text is generated one token at a time (autoregression).

Every LLM is Input → Processor → Output. Text goes in; a probability for the next word comes out.

It only ever predicts the next word

An LLM does not "know" facts the way a database does.

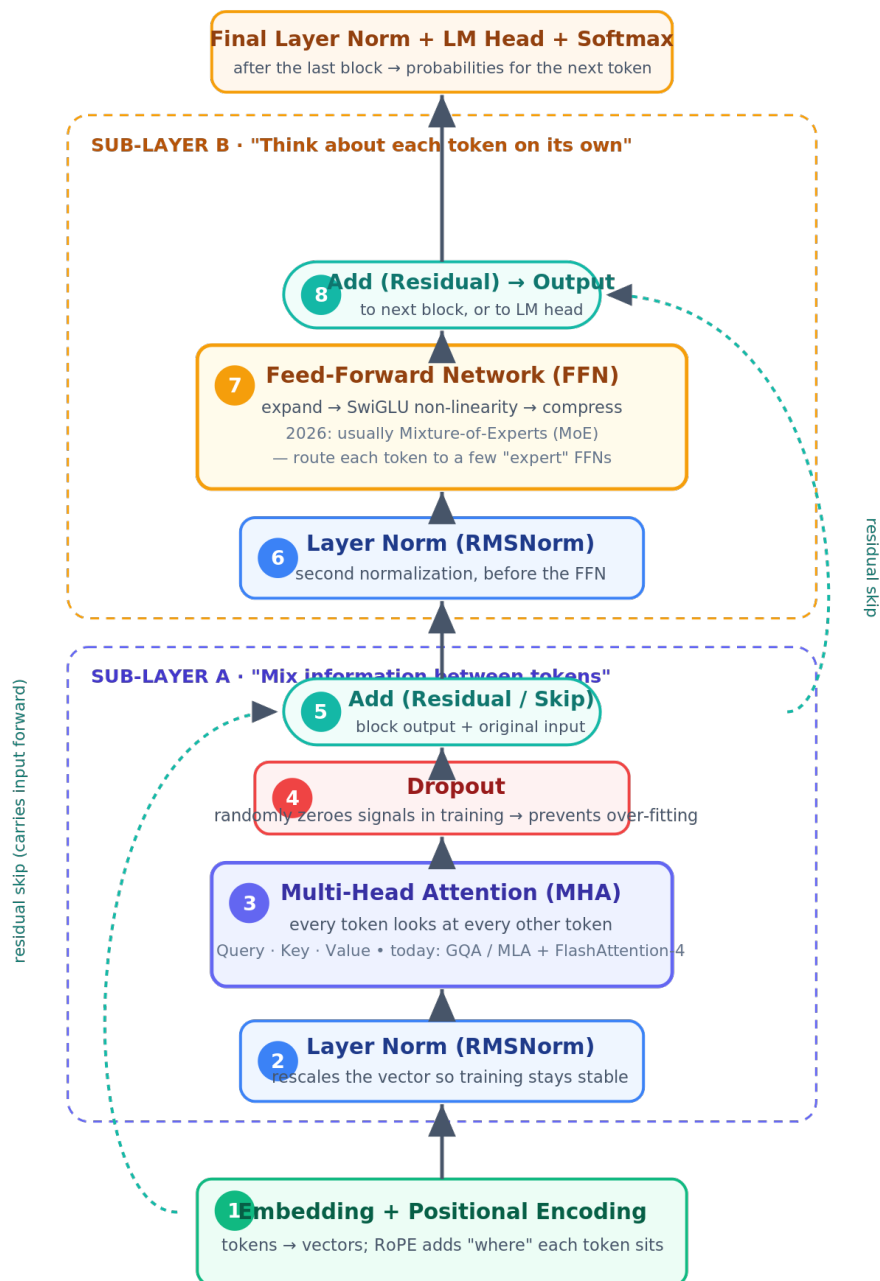
Given "weak results ... board ... replace," the most plausible next token is "CEO." That single move — predict, append, repeat — is everything it does. Drafting, summarising, coding: all the same trick at scale.

Why it matters: that is also why models can hallucinate — they predict plausibility, not truth.

One block = 8 modules

Inside One Transformer Block — The 8 Modules

Modern "pre-norm" decoder block (the 2025–2026 standard). Data flows bottom → top.



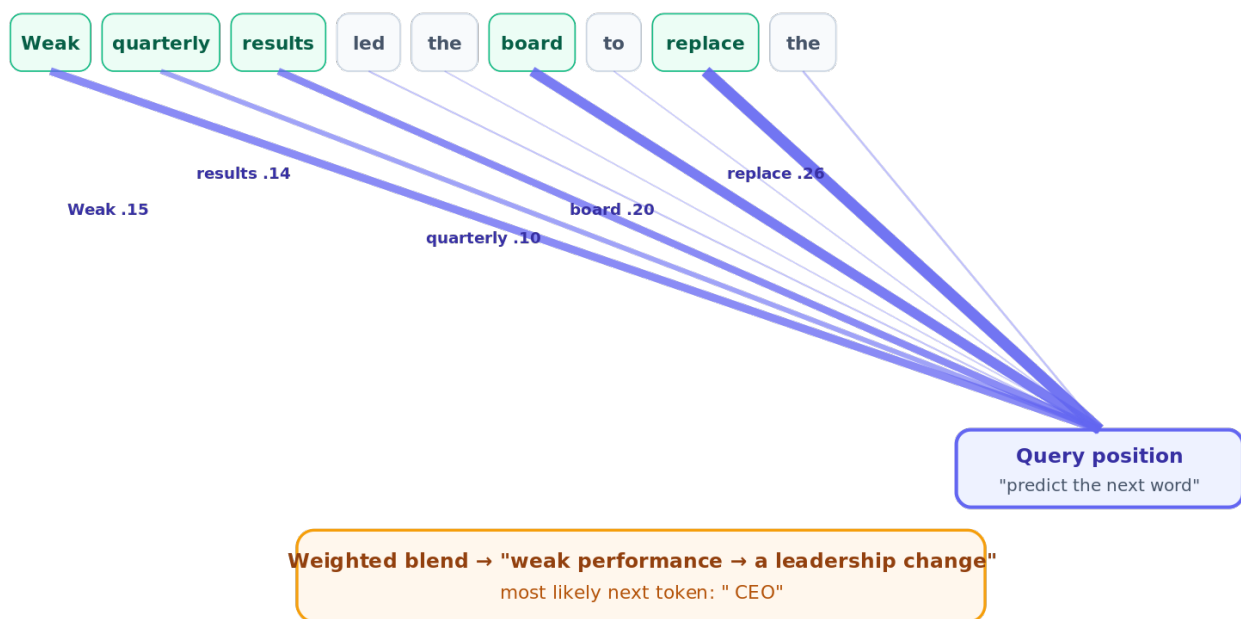
Embedding · LayerNorm · Attention · Dropout · Residual · LayerNorm · Feed-Forward · Residual.

HOW IT DECIDES

Attention weighs what matters

How Attention Decides the Next Word

The position about to be predicted (the Query) scores every earlier token (the Keys). Thicker line = more attention.



"replace", "board", "Weak" and "results" win the most attention — together they point to a leadership change.

The blank looks back at every word and weights them — "replace, board, weak, results" win, so it lands on "CEO."

Same idea, far more efficient

- **Mixture-of-Experts (MoE)**
huge capacity, a fraction of the compute per word
- **RoPE positions**
context windows of 1M+ tokens
- **GQA / MLA + FlashAttention-4**
faster, cheaper inference
- **RMSNorm + SwiGLU**
leaner normalization and activations

The throughline: more capability per dollar.

From architecture to execution

Scoping

Knowing the model guesses tells you exactly where humans must stay in the loop.

Cost

MoE reshapes the unit economics of every AI engagement you price and deliver.

Talent

Your teams don't invent architectures — they wield them. This is the level to lead from.

If you can read one diagram, you can lead any AI conversation.

Full write-up and the matrix-level math deep-dive are linked in the comments.

What AI question do your clients keep asking?

Drop it below — I'll tackle a few.

Follow for plain-English AI-for-delivery breakdowns.